

Statistical Classification of Remote Sensing Satellite Constellations

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Abstract—In recent years, heterogeneous satellite constellations that use inter-satellite links (ISLs) between two different functional constellations, a remote sensing satellite constellation (RSSC) and a relay communication satellite constellation (RCSC) dedicated to command delivery and relay data back to Earth, have attracted considerable interest. The design of general-purpose RCSCs that serve a range of RSSCs is complicated by the many different RSSCs that have been deployed to date and the effort required to test the performance of a candidate RCSC against each one. Accordingly, we have sought to determine whether a pattern in RSSCs exists that would allow us to represent the greater population by a much smaller set of representative classes.

Based on a statistical analysis of NORAD Two-Line-Elements (TLEs) for 34 RSSCs and a thorough review of their missions, we propose a nine-class solution based upon orbital inclination (three categories) and the number of planes and different altitudes (three categories) although only seven of these classes are actually populated by actual RSSCs. The actual number of planes, orbital altitudes, and orbital inclinations are represented by simple distributions. The results allow the performance of RCSCs to be tested against a broad and representative set of RSSCs with far less effort than testing against an exhaustive set and far more confidently than against a random set. A comprehensive survey of the constellation configurations for RSSC missions, currently operational and under-development, are provided.

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1. INTRODUCTION

Satellite constellations for Earth Observations (EO) missions were the first successful implementation of distributed Earth satellite systems that respond to a functional need such as global coverage with low revisit time (RT) [1]. System response time (SRT), which is the interval between submitting a user observation request and receiving the image

product, is the key metric in assessing the timeliness of EO imagery [2]. For disaster monitoring and national security applications, SRT is a more important metric of performance than RT, where the time it takes for the final distribution of the data to the end-user has much more impact on overall mission performance, rather than the time interval between target revisits [3].

SRT can be significantly reduced by using heterogeneous satellite constellations that use intersatellite links (ISLs) between two different functional constellations, a remote sensing satellite constellation (RSSC) and a relay communication satellite constellation (RCSC) dedicated to command delivery and relay data back to Earth [3]. These heterogeneous constellations have been proposed in previous works to explore the potential enhancement of the system performance [2], [4]–[6] or to evaluate the network performances by comparing candidate RCSCs for servicing RSSCs [7], [8]. In most cases, multiple candidate configurations of RCSCs may be appropriate for servicing the same RSSC. Traditionally, to minimize deployment cost, these configurations are selected to minimize the number of relay satellites given the coverage requirements of the RSSC configuration. On the other hand, when designing such heterogeneous constellations, the constellation's network performance (i.e., the quality of ISLs) should become a criterion for selecting a RCSC [9]. Therefore, the best configuration of a heterogeneous system requires studying several constellation combinations.

The heterogeneous system configuration and the constellation details have a significant impact on system availability. The availabilities are dependent on the RSSC & RCSC characteristics, e.g., orbital altitudes, inclinations, number of orbital planes & types, number of satellites per plane, etc. Previous progress in this area has been hampered because designing and evaluating the performance of these heterogeneous systems are difficult due to the large size of the solution space and the input variables. Therefore, previous works use two ways to represent the RSSC in LEOs, a celestial sphere at specific altitude as a lower limit for the existing LEO satellites [4], [5], which is unrealistic representation, or a random configuration such as a constellation of 5 satellites in 3 planes on Sun-synchronous orbit (SSO) [2]. This requires a unified classification of the RSSCs to get representative and realistic constellation

configurations when designing a RCSC for servicing RSSCs in LEO.

The related previous work and survey papers have focused on categorizing and classifying remote sensing satellites based on their mission application or the remote sensor type [10]–[12]. An extensive and compendium survey in [13] has covered the EO missions on a global scale and introduces a survey of these missions and their sensors (past, present, and future). However, they did not introduce a classification for RSSCs configurations. Ulybyshev presented a short historical survey of the constellation design methods of global continuous coverage using circular and elliptic orbits [14]. He classified the satellite constellations into two major types, homogeneous and nonhomogeneous. Homogeneous constellations are that included a number of orbit planes with an equal number of satellites in each plane and the same inclination and altitude while he classified the nonhomogeneous satellite constellations into two subgroups: 1) constellations in different inclinations and equal altitudes 2) constellations in different inclinations and altitudes. This survey was based on the traditional constellation design methods such as Streets Of Coverage (SOC) [15]–[18], Walker [19], Draim [20], and Flower constellations [21], which are not applicable to the remote sensing missions.

TS Kelso's CelesTrak website [22] provides Two Line Elements (TLEs) [23] for use with the Simplified General Perturbations - 4 (SGP4) orbital propagator [24]. Satellites are grouped with common purposes e.g., weather, Earth resources, disaster monitoring, and the Planet and Spire constellations (due to their sizes). These TLEs are classified by common mission rather than orbit or sensor types and there is no classification or grouping to the RSSCs based on the configuration type.

The objective of this paper is to provide representative configurations of the RSSCs instead of using random or unrealistic configurations as used before. These configurations are useful for the current and future simulation-based studies in designing and evaluating the performances of the heterogeneous satellite constellations by proposing realistic and representative RSSCs configurations. Based on a statistical analysis of the NORAD TLEs for 34 RSSCs and a thorough review of their missions, we propose nine representative classes that allow the performance of RCSCs to be broadly assessed with far less effort than testing against an exhaustive set. These classes are based upon orbital inclination (three categories) and the number of planes & different altitudes (three categories) although only seven of these classes are actually populated by actual RSSCs. The actual number of planes, orbital altitudes, and orbital inclinations are represented by simple distributions.

In this paper, Section 2 presents a review of what already known about the constellation configurations and orbit types used for RSSC missions by introducing a representative list of the RSSC missions. The data sources and the methodology used in the statistical analysis of these data are discussed in Section 3. In Section 4, we discuss the main observations in

the results we have got from the statistical analysis of the collected data and introduce the proposed representative classes for RSSCs configurations. Finally, Section 5 concludes the paper.

2. A REPRESENTATIVE LIST OF RSSC MISSIONS

Traditional satellite systems for EO missions have some key limitations that can impact their utility e.g. 1) Most current systems consist of a single satellite, or a small constellation with 3-5 assets, resulting in limited coverage and the revisit performance will be of the order of days 2) The majority of imaging satellites are in high inclination polar orbits which are optimized for global coverage at the expense of revisit times 3) The optical satellites are generally in SSOs with fixed imaging local time of ~10:30 am to maximize illumination conditions and minimize cloud cover, resulting in minimal coverage at different times of the day 4) Synthetic-aperture radar (SAR) satellites are generally in SSOs with a fixed imaging local time of ~06:00 am as the satellites' solar panels can always see the Sun, without being shadowed by the Earth [25].

These limitations can be overcome by a moderately sized constellation of small satellites in different orbit types and inclinations, which can provide a number of advantages including:

1. Enable activity monitoring and change detection with near persistent observation being potentially achievable using large constellations that can improve the temporal resolution (revisit time).
2. Easy replacement of a satellite within a constellation due to the relatively low costs of a single satellite.
3. Soft degradation of the system performance caused by the malfunction of one satellite.

Nevertheless, the constellation design process for RSSC missions is a complex trade-off between [26]:

1. Overall mission cost, which is a function of launch cost, the total number of satellites, number of planes, and the satellite size and mass.
2. Providing high-resolution images.
3. Providing fast imaging delivery.

In this section, a representative list of different RSSC configurations in different missions and applications is introduced to show how the constellation and orbital parameters are selected to overcome the mission limitations and to achieve the mission requirements. We introduce some examples of RSSCs based on the sensor type, which can be classified into 3 categories: SAR, optical, and weather & scientific constellations. The following examples of RSSC missions are given in order to show that different remote sensing applications need different approaches for cost-effective missions and different constellation configurations and parameters. Moreover, they reflect what we have already known about the general trends of configurations and orbital parameters for RSSCs.

SAR Constellations

RADARSAT

Synthetic-aperture (SAR) sensors, with huge mass and power requirements, have been flown by large satellites such as the RADARSAT satellite constellation (RCM), Canada's new generation of EO satellites. RCM is a three-spacecraft fleet operated by the Canadian Space Agency launched on 12 June 2019. The primary goal of RCM is to provide continuous C-band SAR data to RADARSAT-2 users, as SAR imagery at a high temporal resolution is required by several users in the Canadian government. Other improvements include more frequent area coverage of Canada and reduced risk of a service interruption. Its SAR sensors have a mass of 400 kg each and a high resolution of 1×3 m. There is also a secondary payload, an Automatic Identification System (AIS) for ships, that will be used independently or in conjunction with the SAR [27].

The primary areas of interest of RCM are the landmass of Canada and its surrounding Arctic, Pacific and Atlantic maritime areas. RCM is a trio EO satellites in one SSO plane spaced apart by 120° at a mean altitude of 593 km at 97.74° inclination. Its 12-day repeat cycle provides a coherent change detection period of 4 days (taking into account three satellites) and optimizes uniformity of repeat coverage to the areas of interest while maintaining global access with each satellite [28]. RCM orbit selection was based on trading off several considerations. First, important considerations for the nominal altitude are power and fuel. Second, secondary considerations when choosing the orbit altitude include ground station (GS) contact time, link margin, and incidence angle variation across the imaged swath. Finally, given an approximate range of altitudes, orbit selection needs to consider the repeat cycle.

ICEYE

Active microwave payloads such as SAR sensors have been typically associated with large and expensive platforms, mainly due to their mass and required power [29]. However, while in markets timeliness is valuable, ICEYE is making trade-offs between providing high-resolution SAR images and fast image delivery. Finland's Aalto University plans to provide data with a constellation of six SAR satellites focused on Earth's Arctic regions and build up to a constellation of 18 space-based radars in collaboration with the European Space Agency (ESA) to offer customers the ability to acquire data on any area of the globe within a couple of hours of a request [30].

Compared to existing SAR constellations, ICEYE with a large number of satellite units distributed on multiple orbital planes (6 SSOs at ~ 500 km altitude) enables shorter response time from data acquisition requests to delivery on the scale of 2-3 hours in the arctic latitudes. To reduce the size and cost of the ICEYE satellites and thus enable this large number of satellites in the constellation, it is necessary to have a small imaging payload, which is an X-band SAR designed by

ICEYE. The instrument is built with several trade-offs focusing mostly on low unit mass and cost so that the high temporal resolution can be realized in a commercially feasible way but with an image resolution of 10×10 m [31].

OptiSAR

The OptiSAR constellation comprises eight tandem pairs of SAR and optical satellites divided into two orbit planes. The four tandem pairs will be equispaced around an orbit plane, where each tandem pair consists of a leading SAR satellite, which uses UrtheCast's SAR-XL technology that provides a dual-band (X-band and L-band) SAR instrument, and a trailing optical satellite that is following approximately 2 minutes behind the SAR satellite. The first orbit plane is a SSO with a 10:30 a.m. equator crossing time that is a commonly used orbit for EO missions, and the second plane is a Mid-Inclination Orbit (MIO) with a $\sim 45^\circ$ inclination. The MIO is used to provide ultra-high revisit in mid-latitude regions of the Earth where the bulk of the world's population resides. Both orbit planes have a satellite altitude of 450 km [33]

Traditionally, the power-hungry nature of SAR sensors combined with their day and night imaging ability have driven SAR satellites to fly in dawn-dusk orbits to maximize power generation and the payload duty cycle. This, however, hasn't been the case for optical satellites due to the less than optimal ground illumination conditions in such orbits. A key feature of the OptiSAR constellation is the cross-cueing capabilities between the SAR and optical satellites which enables the system to have a wide swath surveillance mode with the SAR satellite. Using onboard processing, it is capable to detect, classify objects of interest (e.g., ships), and then automatically task the optical satellite to take a high-resolution image or video of the selected objects [33].

Optical Constellations

RapidEye

RapidEye is a commercial multispectral EO mission that includes a constellation of five minisatellites (satellite launch mass ~ 156 kg) that has been launched on a single Russian Dnepr rocket from the Baikonur Cosmodrome in Kazakhstan in August of 2008 [34]. They are deployed in orbits at an altitude of 630 km. The satellites are placed equally spaced in a single SSO to ensure consistent imaging conditions and short RT. The satellites follow each other in their orbital plane at about 19 min intervals. The constellation approach in a single orbital plane permits a cumulative swath to be built up (the spacecraft view adjacent regions of the ground, with image capture times, separated by only a few minutes). The RT of one day can be obtained anywhere in the world ($\pm 70^\circ$ latitudes) with body pointing techniques. The average coverage repeat period over mid-latitude regions (e.g. Europe and North America) is 5.5 days at nadir. The RapidEye system can access any area on Earth within one day and cover the entire agricultural areas of North America and Europe within five days. The uplink and downlink capability are

provided by reception stations located at Svalbard in Norway and operated by Kongsberg Satellite Services (KSAT) [35].

PlanetScope (PS)

The standardized CubeSat concept [36] has offered a cost-effective solution to revolutionize the EO potential. This concept uses compact and light-weight (<1.33 kg) single-unit (1U; $10\times10\times11.35$ cm) building blocks as the modular basis for forming larger satellite configurations (i.e., 3U, 6U, 12U) for a variety of application fields [37]. As CubeSats are comprised largely of commercial off-the-shelf components and deployed as a secondary payload on reusable rockets, the changing economics of EO has made it feasible to launch CubeSats in flocks at an unprecedented rate.

PS is a satellite constellation consisting of 170+ CubeSats ("Doves") operated by Planet Labs, an American commercial Earth-imaging company based in San Francisco, CA. Their goal is to image the entirety of the planet daily to monitor changes and pinpoint trends. Each Dove satellite is a 3U CubeSat. The number of CubeSats in this constellation is still increasing. The majority of CubeSats are in SSOs (~ 475 km altitude) with a midmorning equatorial overpass time (9:30 – 11:30 a.m., local solar time) and a nadir GSD between 3.5 – 4 m. Smaller flocks of PS satellites still operate in the International Space Station (ISS) orbit (~ 400 km altitude), which results in a variable equatorial overpass time and a GSD of ~ 3 m. Doves form the largest satellite constellation in the world that provides a complete image of Earth once per day. The constellation is constantly "on" and does not require ordering or acquisition planning [38].

DigitalGlobe (DG)

DG is a Maxar Technologies company headquartered in Westminster, Colorado, has five very high-resolution Earth-imaging satellites (GeoEye-1, WorldView-1, WorldView-2, WorldView-3, WorldView-4). With this fleet of agile satellites and large telescopes for very high resolution (VHR) imagery, DG offers today the sharpest imagery worldwide (up to 30 cm GSD with WorldView-3 and WorldView-4 launched in November 2016) [39]. The constellation consists of five satellites distributed nonuniformly on four different SSO planes. In January 2019, Maxar reported that WorldView-4 satellite experienced a failure in its CMGs (Control Moment Gyros), preventing the satellite from collecting imagery due to the loss of an axis of stability [40]. Each satellite in the constellation has its own plane with different altitudes with a non-uniform phasing crossing the Equator.

WorldView-Legion is DG's next generation of EO satellites that consists of six satellites planned to be launched in 2021 into a mix of SSO and mid-latitude orbits. These satellites will replace imaging capability currently provided by WorldView-1, WorldView-2, and GeoEye-1. The initial constellation is to be orbited with two satellites going to SSOs and four to 45° orbits [41].

Unlike PS constellation that can provide a complete image of Earth once per day, DG constellation can capture the earth's landmass in 60 days. However, DG is the only commercial company with a 30-cm product on the market to extract detailed information from phenomena that the smaller satellites can image only vaguely.

Weather and Scientific Constellations

FORMOSAT-3/COSMIC

FORMOSAT-3/Constellation Observing System for Meteorology, Ionosphere, and Climate (COSMIC) is a joint mission of Taiwan and the U.S. The mission consists of six identical LEO satellites. The constellation was launched on April 15, 2006, into the same orbit plane of a designated circular parking orbit, at an altitude of 516 km. This was being followed by a 13-months constellation deployment/distribution sequence until the distribution of six orbit planes was achieved. Currently, each satellite has its own plane non-uniformly distributed around the Equator (orbits are phased $\sim 30^\circ$ apart in ascending node and 52.5° apart in the argument of latitude) at ~ 793 km altitude and 72° inclination [42].

Until the launch of FormoSat-3/COSMIC, there had been little GPS radio occultation (GPS-RO) data, which is a remote sensing technique that relies on the change detection of GPS radio signals as it passes through Earth's atmosphere for measuring its physical properties and performing weather forecasting and monitoring climate change. The major source had been the German CHAMP satellite (launch July 15, 2000), which has delivered 150-200 profiles per day globally. COSMIC's six satellites have generated a total of more than 2,500 GPS-RO data per day. The retrieved RO data have been assimilated into numerical weather prediction models by many major weather forecast centers and research institutes [42].

NOAA and Taiwan's National Space Organization (NSPO) intend to jointly develop and launch COSMIC-2, a high-reliability next generation follow-on system. FormoSat-7/COSMIC-2 will provide the next generation of global navigation satellite system GNSS-RO data to users. This constellation is expected to be a much-improved constellation system consisting of twelve satellites, six satellites at 72° inclination, and six satellites at 24° inclination (currently operational), which will enhance observations in the equatorial region over what is currently being collected with the COSMIC mission. This constellation configuration was chosen because it provides the most uniform global coverage to collect a large amount of atmospheric and ionospheric data. This constellation will produce more than 8,000 soundings per day, compared to the approximate 2,500 soundings per day currently produced by COSMIC due to COSMIC-2's ability to track three navigation systems' signals versus COSMIC's ability to track one [43].

NASA's CYGNSS

Cyclone Global Navigation Satellite System (CYGNSS) is a constellation of eight microsatellites launched into low inclination LEO over the tropics on December 15, 2016, non-uniform distributed on a circular inclined orbit at altitude 450 km. The 34° orbit inclination was selected to maximize the dwell time over latitudes at which hurricanes are most likely to occur. This configuration will make frequent measurements of ocean surface winds in the tropics, with a primary objective of monitoring the location, intensity, size, and development of tropical cyclones. Successive spacecraft in the constellation observed Enawo over a period of several hours just before it made landfall on Madagascar on March 6, 2017 [44].

CICERO

Community Initiative for Continuing Earth Radio Occultation (CICERO) constellation is developed by GeoOptics Inc. of Pasadena, CA. The precise number of satellites (~26 of 6U CubeSats in total) and the timeline for their deployment will depend on customer demands. The first four operational RO satellites are currently on four different SSOs at altitudes 493 and 497 km. The next phase will comprise of ten satellites and occupy circular orbits at 500-700 km at inclinations above 65°. These cells are collecting GPS and GLONASS RO profiles to return 500 - 1,000 profiles per day. Later launches will introduce cells at lower inclinations to densify coverage of the mid-latitudes and tropics. They will add Galileo and BeiDou tracks, doubling the per-cell data return. The constellation orbital parameters are optimized to maximize the number and the distribution of atmospheric soundings that maximize the global distribution of occultation events [45].

To sum up, the design process of the RSSCs starts from establishing mission objectives and constraints according to mission requirements. Therefore, objectives and constraints of remote sensing missions are the main drivers of development remote sensing sensors, evolving the missions' metrics of performance, and selection the constellation orbital configurations & the number of satellites based on simulation methods and orbital mechanics laws [3]. The selection of orbital parameters for the constellation mission is the most important factor for the fulfillment of the space mission requirements. Practically, a straightforward approach for the selection of these parameters does not exist. Therefore, it is necessary to follow a complex process that requires trade-offs among the different design parameters and the corresponding metrics of performance such as coverage, revisit time, sensor resolutions (spatial, spectral, and temporal), the capacity of data collected, swath width, spectral ranges of interest for every application, and system response time. Consequently, there are various different configurations of RSSCs.

3. METHODOLOGY

Data Sources

We performed a statistical analysis of the data sets of the Weather & Earth Resources Satellites groups provided from CelesTrak for the remote sensing satellites in LEO. This analysis is based on the TLEs data available mainly from the CelesTrak website with the help of the data available from the Space Track website [46]. Then, The satellites that belong to a constellation mission were grouped using different methods 1) data analysis of the orbital elements extracted from TLE data sets 2) orbit visualization of these data sets using AGI's STK standard object database with the help of Cesium, which is available on CelesTrak website 3) literature survey of the RSSCs missions.

The TLE data for CelesTrak comes from Space Track, although on CelesTrak, they perform a variety of additional checks and provide supplemental TLEs that Space Track does not. In general, CelesTrak is easier to use (e.g., not need to register and log in) and does not place restrictions on the use of the data. Additionally, they provide an improved Satellite Catalog (SATCAT), which includes data not included on Space Track, such as operational status, Radar Cross Section (RCS) values (up until recent years), along with more timely and accurate identifications of new launches. Moreover, they provide groups of satellites for common purposes like the Earth Resources list, which contains TLE data for a large group of operational remote sensing satellites, weather, disaster monitoring, and the Planet and Spire constellations (due to their sizes) — none of which is on Space Track.

Statistical Analysis Reductions

We collected the TLE files of all remote sensing satellites from CelesTrak groups that are currently operational. Using Matlab, the data string of NORAD TLEs of these groups were converted into a data structure in order to get all the data provided in TLEs for each satellite. The following information was derived from TLEs:

1. Keplerian orbital elements including semi-major axis a from satellite mean motion, inclination i , eccentricity e , right ascension of ascending node (RAAN), the argument of perigee ω , and mean anomaly)
2. Object ID (satellite name)
3. Launch Year
4. Launch Number

Statistical analysis of the remote sensing satellites in each group gives us some general and commonly-known information about the remote sensing satellites, e.g., most of the satellites are on circular or near-circular orbits and the majority of these satellites are on high inclinations, definitely, SSOs. Since this information does not reveal anything about the constellation configurations, grouping satellites into

constellations is required for the statistical analysis of the RSSCs.

In order to group these satellites, we have to know which ones are in which constellation. In many cases, the satellite's name (e.g. FLOCK or LEMUR) can be used to tell which constellation they are in, but that's not always the case. Therefore, a survey study is required on each constellation mission to confirm that all of these satellites are in the same constellation and add the missing satellites with different names. For example, the GeoEye-1 satellite belongs to the DG satellite constellation while the other 4 satellites are WorldView 1 – 4. We also used the launch year and launch number in the year of all of these satellites to distinguish whether satellites are in the same planes or not. For example, the five RapidEye satellites have the same launch number 40 with the same year 2008 thus, they are on the same plane. Note that while it is typical to maintain satellites from a particular launch in the same orbital plane that is not always the case.

The best way to test whether satellites are in the same plane is to calculate the RAAN from the TLE orbital data. As an example, the data from Disaster Monitoring Constellation (DMC) satellites indicates that RAAN values of DMC3-FM1, DMC3-FM2, and DMC3-FM3 satellites are slightly different while the three satellites are in the same orbit (almost the same altitude and inclination), refer to Table 1. In such cases, we use the AGI STK standard object database to visualize these satellites to confirm the data we have. Figure 1 shows the three DMC3 satellites which are almost in virtually the same plane, but not exactly the same. Here, where FM1, FM2, and FM3's orbits are in red, white, and green, respectively. The RAAN is quite close for FM1 and FM2, but off a bit for FM3, which are in accordance with values in Table 1.

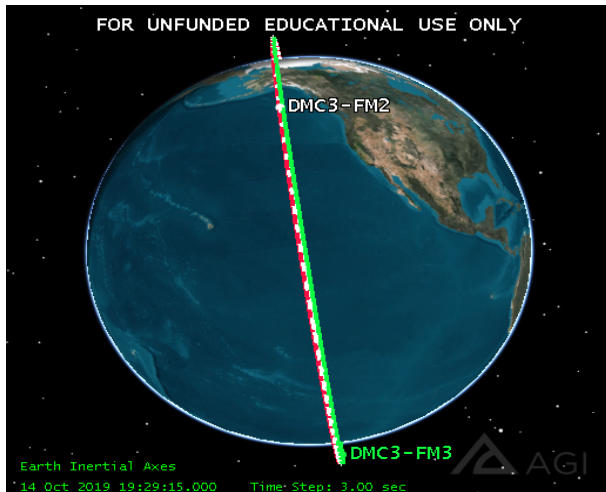


Figure 1 – Visualization of DMC3 satellite constellation.

Table 1 – DMC3 satellite data from TLEs.

Satellite name	DMC3-FM1	DMC3-FM2	DMC3-FM3
Launch year	2015	2015	2015
Launch number	32	32	32
<i>a</i> (km)	7025.9117	7025.897	7025.927
<i>e</i> (deg)	0.0015855	0.0012957	0.0016872
Mean altitude (km)	647.771727	647.757015	647.786953
<i>i</i> (deg)	98.0089	98.0108	98.0096
RAAN (deg)	26.3471	26.5179	27.2491

After processing and analysis of the TLE data from CelesTrak, we created satellite groups of different RSSCs that are currently operational. Some other under development constellations were added from a literature survey of RSSC missions such as CICERO, Vivid-i, OptiSAR, NovaSAR, XpressSAR, MicroSAR, and ICEYE to identify the configurations of 34 RSSCs with 348 satellites, which represent the majority of the RSSCs to date and near future. We grouped the remote sensing satellites into constellations using different sources: 1) the orbital elements data extracted from TLE data sets, 2) orbit visualization of these data sets using AGI's STK standard object database and Cesium, which is available on CelesTrak website, and 3) literature survey of the remote sensing constellation missions.

Statistical analysis of the constellation parameters of these constellations is performed. In the analysis of these constellation configurations, we obtained the histograms of the total number of satellites, number of planes, average orbital altitude, average eccentricity, average inclination, and the uniformity of satellite distributions for each constellation. Then, we generated a correlation matrix between these variables to see patterns that help in classification the RSSCs.

4. RESULTS AND DISCUSSION

Analysis of Satellites in CelesTrak Groups

In this section, histograms for all of the LEO satellites in separate lists of CelesTrak including weather, Earth resources, disaster monitoring, and the Planet and Spire constellations are presented. These histograms represent the distribution of satellites' orbital parameters, altitude, eccentricity, RAAN, and inclination. This analysis was performed based on the TLE data obtained on July 28, 2019. We analyzed these satellite groups to check if there is any pattern among constellation parameters that may help in the RSSC classification.

Figure 2 shows the histograms of 10 satellites listed in the disaster monitoring group. Figure 3 shows 162 satellites listed in the Earth resources group. Orbits of these satellites are circular or near-circular. The majority of these satellites are using SSOs in the altitude range from 500 km to 800 km. Very few of these satellites are using inclined orbits such as the Chinese commercial satellites, Zhuhai-1 01 and Zhuhai-1 02, which are using 43° inclination to provide data services for areas including agriculture, land and water resources, environmental protection and transport in China [47]. 237 satellites listed in the Planet group are shown in Figure

4. This group includes all the FLOCK satellites in the SSO besides RapidEye and SKYSAT constellations. Figure 5 shows the histograms of 84 satellites listed in the Spire group. These satellites are using circular orbits with multiple different altitudes and inclinations including the ISS and near-equatorial inclinations. Figure 6 shows the histograms of 56 weather satellites listed in both weather and NOAA groups on CeresTrak. The majority of these satellites are using high inclination ($\sim 100^\circ$), near-circular orbits, and high LEO altitude ranges from 780 km to 850 km and from 1,446 to 1,520 km.

Statistical analysis of the remote sensing satellites in each group gives us some general and commonly known information about remote sensing satellites such as SSO is the most widely used orbit type and most of them are circular or near-circular orbits. Although we could not use these histograms to classify the configurations of RSSCs, they helped us to group the remote sensing satellites into constellations.

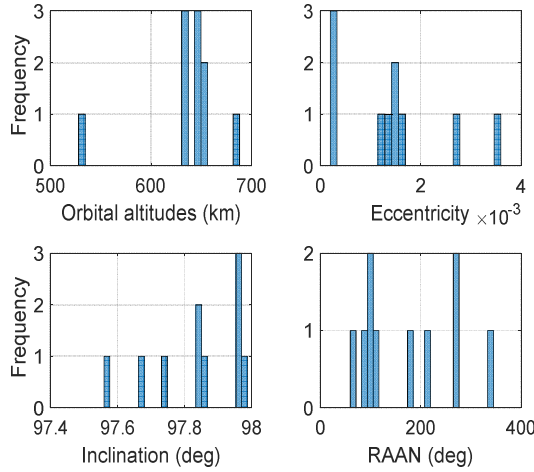


Figure 2 – Histogram of all satellites in the Disaster Monitoring group on CeresTrak.

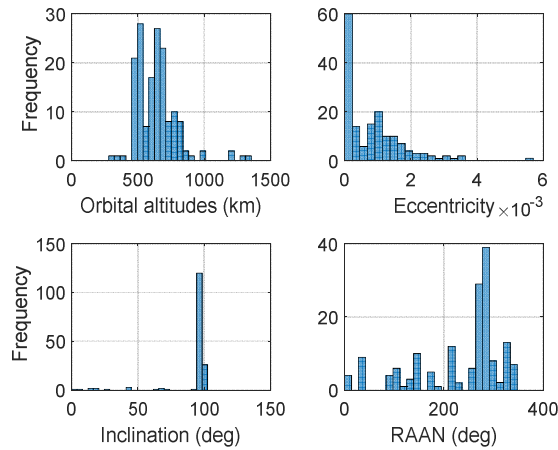


Figure 3 – Histogram of all satellites in the Earth resources group on CeresTrak.

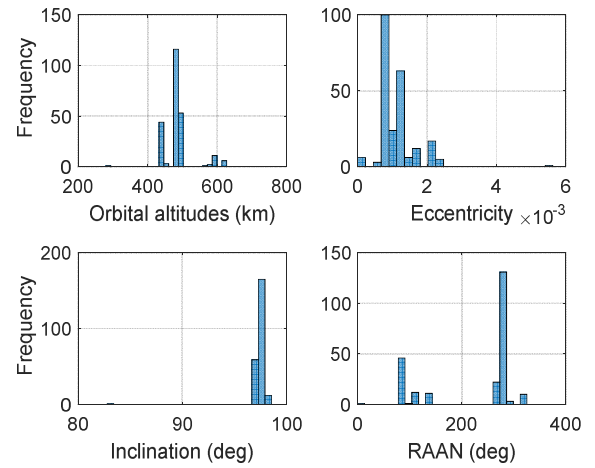


Figure 4 – Histogram of all satellites in the Planet group on CeresTrak.

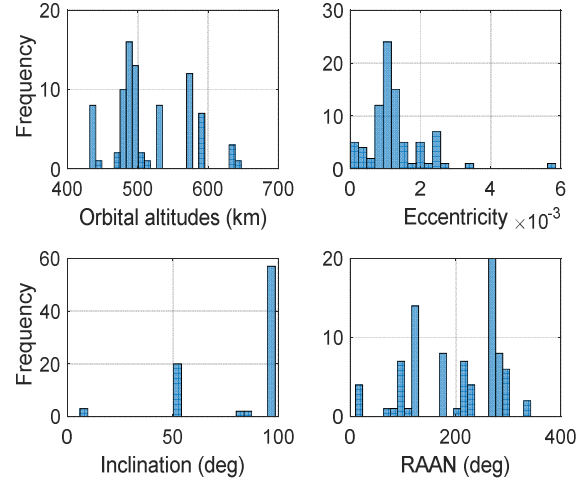


Figure 5 – Histogram of all satellites in the Spire group on CeresTrak.

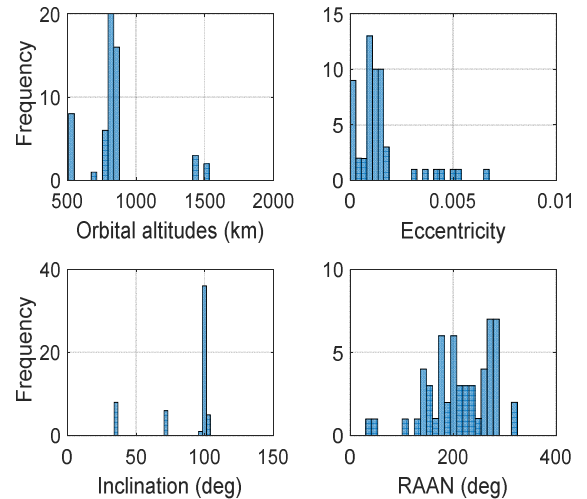


Figure 6 – Histogram of all satellites in Weather and NOAA groups on CeresTrak.

Analysis & Classification of RSSCs

A comprehensive survey of the constellation configurations for RSSC missions, currently operational and under-development, are provided. This survey is based on the analysis of the orbital parameters provided in the NORAD TLEs with the help of two other methods for the sake of verifications, interactive visualization provided by the CelesTrak website, and STK standard object database. From the literature survey of these constellation missions, we classified these constellations into four groups based on sensor type. Table 2 summarizes the weather monitoring constellations. Table 3 summarizes RSSCs that are using optical sensors. Table 4 summarizes RSSCs that are using SAR sensors and Table 5 for satellite constellations that have both optical and SAR sensors. These tables summarize the information on the constellation configuration, their operational status, and the available locations of Payload Ground Segment (PGS), which is responsible for the management and receiving the payload data of the constellation satellites.

The constellation parameters of the constellation groups are collected from Tables 2-5. Figure 7 shows the histograms of these parameters i.e., the total number of satellites, the number of planes, the average orbital altitude, average eccentricity, average inclination, and the uniformity of satellite distributions in each constellation. Note that the uniformity distribution in this figure, 1 indicates uniformity of the satellites' distribution in a constellation while 0 refers to non-uniformity distribution. From these histograms, we obtained the major RSSC parameters and their range of values as:

1. The number of planes: from 1 to 6 planes.
2. The total number of satellites: from 4 to 20 satellites.
3. Orbital altitude: from 400 km to 800 km.
4. Inclinations: from 97° to 100° (based on SSO altitude), ISS inclination (~51°), mid-inclinations such as (34° and 45°), and finally 70°.

The correlation coefficients between each pair of the parameters shown in Figure 7 is presented in Figure 8 in order to find out any patterns among these parameters that help in RSSC classification. It is concluded that these variables are almost independent. The reason is that for each RSSC mission, the objectives and constraints are the main drivers in constellation design process and every constellation mission has its specific objectives and constraints and there is no straightforward approach for selecting their constellation parameters.

From these combinations of observations, we propose a heuristic classification of the RSSCs. This classification introduces a nine-class solution based upon orbital inclination including SSO, inclined orbit and hybrid and the number of planes and different altitudes including same

plane-same altitude, different plane-same altitude and different planes-different altitude although only seven of these classes are populated by actual RSSCs. Table 6 summarizes these solutions given representative constellation missions for each class.

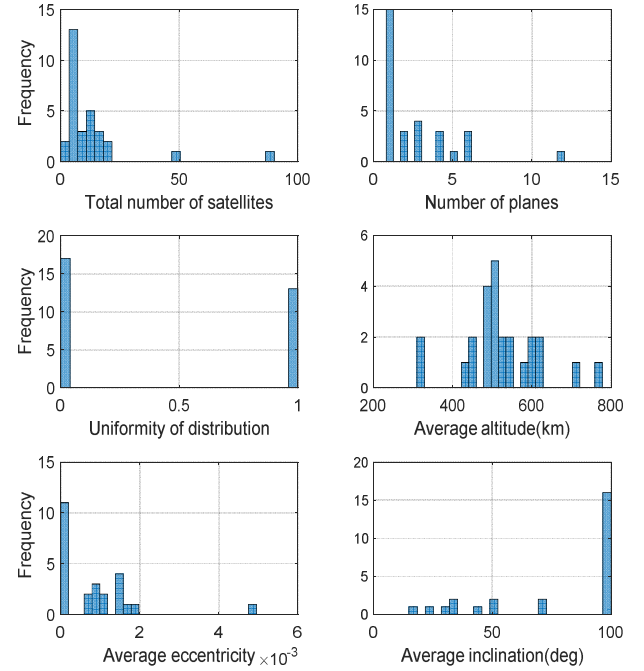


Figure 7 –Histograms of selected parameters in RSSCs groups.

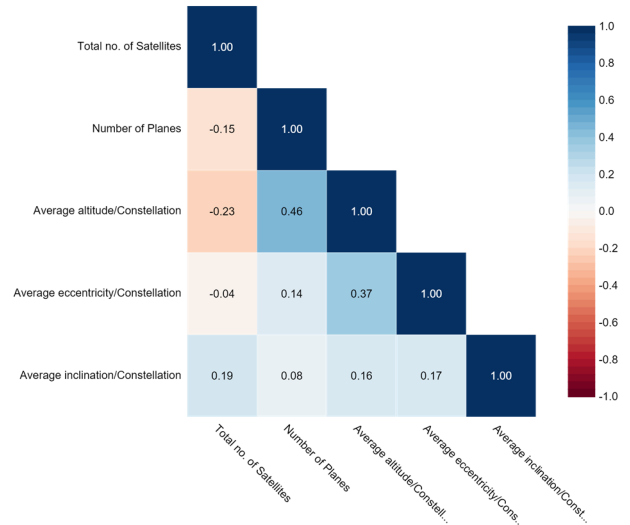


Figure 8 – Correlation matrix of the constellation group parameters

Table 2 – Weather monitoring constellations.

Mission	# Sats	# Orbits & type	Constellation configuration comments	Operational status	PGS
FormoSat-3/ COSMIC (FS-3/C)	6	6 circular inclined	6 non-uniform distributed orbit planes. All satellites have the same altitude 793 km and inclination 72°.	2006 Operational	ground station network (consists of 4 GSs)
FormoSat-7 / COSMIC-2 (FS-7/C2)	12	12 near- circular inclined orbits	Two sets of six satellites: the first set, COSMIC-2A, operates in low-inclination orbits with 24° inclination and altitude of ~ 520-550 km. The second set, COSMIC-2B, would operate in highly inclined orbits (72°) and altitude of ~ 720 km for replacement with the original FormoSat-3/COSMIC constellation that launched in 2006.	Under development	Planned 8 equatorial GS
Lemur Nanosatellite Constellation of Spire Global	57	SSO, ISS, and various circular inclined orbits	Non-uniform distribution. Using many CubeSats at many different orbits for reducing the revisit time.	Operational	Distributed GSs in China
NASA's CYGNSS (Cyclone Global Navigation Satellite System)	8	Circular inclined single plane.	Non-uniform satellite distribution. Satellites have the same altitude (520 km) and the same inclination (35°) but crossing the equator at different RAAN.	Operational	--
CICERO By GeoOptics	26	SSO and mixture of inclined orbits	Non-uniform distribution. The precise number of satellites and the timeline for their deployment will depend on customer demands. The first two operational RO satellites are currently on two different SSOs at altitudes 493 and 497 km. The first 14 CICERO satellites will occupy circular orbits at 500-700 km at inclinations above 65°. Later launches will introduce cells at lower inclinations to densify coverage of the mid-latitudes and tropics.	Under development over the following three years with 2 operational satellites.	Proposed a worldwide license for universal data access
NOAA	20	20	Polar orbits. Each satellite has its own orbit	Operational	Svalbard
NASA TROPICS [48]	6	3 inclined orbits	Six 6.0 kg 3U CubeSats spread across three 550 km altitude and 30° inclination orbital planes.	Launch readiness is projected for late 2019	--

Table 3 – Optical constellations.

Mission	# Sats	# Orbits & type	Constellation configuration comments	Operational status	PGS
RapidEye	5	1 & SSO	Uniformly distributed on a single SSO at the same altitude (627.3 km) and same inclination (97.8°)	2008 Operational	Svalbard 78° N Norway
DMC	8	6 & SSO	Non-uniform distributed. BEIJING 1 is the remaining satellite of 5, which formed the DMC first generation (DMC-1) at altitude 687 km and inclination 97.9°. Second-generation (DMC-2) in orbit since mid-2009. Deimos-1 and UK-DMC-2 are on different SSOs at the same altitude 655 km and the same inclination 97.8°. Nigeria-2 and NigeriaSat-X were launched in 2011 into orbits with slight differences in altitude & inclination. Three third-generation satellites, (DMC3-FM1, FM2, and FM3) are uniformly distributed in the same orbit 120° apart at altitude 648 km and inclination 98°.	8 Operational out of 12 launched satellites (2005-2015)	Ground Station Network
Flock 1C	11	1 & SSO	Non-uniform distributed on the same orbit (597 km altitude and 97.9° inclination with slight differences in RAAN)	Only 5 satellites are operational	Planet Lab Inc. ground station network
Flock 3P	88	1 & SSO	Nearly uniform distributed. Same altitude (489.7 km) and same inclination (97.45°).	Operational Since 2017	
Flock 3M	4	1 & SSO	Non-uniform distributed. Same altitude (506 km) and same inclination (97.35°).	Operational Since 2017	
Flock 3P'	4	1 & SSO	Non-uniform distributed. Same altitude (496.2 km) and same inclination (97.53°).	Operational Since 2018	
Flock 2K	47	1 & SSO	Nearly uniform distributed. Same altitude (454.25 km) and same inclination (96.98°).	Operational Since 2017	
Flock 2e	12	1 & ISS	Non-uniform distributed. Same altitude (318 km) and same inclination (51.6°).	Decayed	
Flock 2e'	20	1 & ISS	Non-uniform distributed. Same altitude (318 km) and same inclination (51.6°).	Decayed	
Flock 2P	12	1 & SSO	Non-uniform distributed. Same altitude (489.8 km) and same inclination (97.4°).	Operational Since 2016	
Flock 3R	15	1 & SSO	Non-uniform distributed. Same altitude (482 km) and same inclination (97.48°).	Operational Since 2018	
SkySat TerraBella	13	4 & SSO	Non-uniform distributed. Skysat-1 on a single orbit at altitude 568.2 km and 97.7° inclination. Skysat-2 on a single orbit at altitude 626 km and 98.4° inclination. SkySat-C series is 11 satellites distributed in two planes. From Skysat-C1 to Skysat-C5 were launched in 2016 and distributed equally on a single SSO at altitude 494 km and 97.35° inclination. From Skysat-C6 to Skysat-C11 were launched in 2017 and distributed equally on another SSO at altitude 498 km and inclination 97.4°.	Operational (2013-2017)	--
Digital-Globe	5	4 & SSO	Non-uniform distributed. 3 satellites each on a separate plane, GEOEye1 (678.5 km altitude), WorldView-1 (489 km altitude), and WorldView-2 (766 km altitude). WorldView-3 and WorldView-4 are equally distributed on a single plane with a slight difference in altitude (611.5 km altitude) between them due to altitude decay of WV-3.	2008-2016 Operational	8 Remote Ground Terminals
Vivid-i	15	3 & SSO	The planned Vivid-i constellation will be launched in batches of five. The first five satellites will be launched in 2019 into SSO of 500 km altitude. CARBONITE-2 was launched in 2018 for a technology demonstration of this constellation at on SSO altitude 505 km.	Announced in May 2017 Under development	Appointed to be Norway's KSAT

Table 4 – SAR constellations.

Mission	# Sats	# Orbits & type	Constellation configuration comments	Operational status	PGS
COSMO-SkyMed	4	1 & SSO Dawn-Dusk	X-band SAR Systems. Uniform distributed on a single SSO at altitude 620 km and inclination 97.9°. During nominal operation, all satellites operate on the same orbital plane with 90° equal-spaced. During Tandem interferometric configuration, they are working in different two orbital planes.	2007-2010 Operational	Multi-center architecture geographically distributed over a wide territory. 3 GSs in Italy, one in Sweden, and one in Spain
COSMO-SkyMed (Second Generation – CSG)	2	1 & SSO Dawn-Dusk	X-band SAR Systems. Same orbit as COSMO-SkyMed. Two satellites that will integrate and finally replace the current operational COSMO-SkyMed satellites will compose CSG.	CSG-1 will be launched in 2018. CSG-2 a year later	
RADARSAT (RCM)	3	1 & SSO Dawn-Dusk	C-band SAR Systems. Equally spaced 3 satellites around the orbit plane at altitude 592.7 km and inclination 97.74° (with a potential to increase the number to six).	Launched in June 2019	Gatineau, Prince Albert, Inuvik and one foreign station in Svalbard
SAR-Lupe	5	5	X-band SAR Systems for the German military to achieve Ground Sampling Distance < 1m. Non-uniform distribution. Each satellite in a single orbit (apogees at 505 km and perigees at 468 km) at inclination 98.2°.	2006-2008 Operational	Single GS in Gelsdorf
NovaSAR	4	SSO or low inclination Equatorial orbit	S-band SAR Systems. Non-uniform distributed. Altitude 580 km and designed for multiple orbits: SSO (LTAN 10:30) or a dawn-dusk orbit (LTAN 6:00) and an Equatorial orbit at a minimum 15° inclination.	Planned to launch in 2018	Astrium network
XpressSAR	4	2 inclined orbits	X-band radar systems. 4 satellites in two orbit planes with a 35° inclination at altitude 425 km.	Planned to launch in 2020	Miami, Florida
MicroSAR System	10	2 inclined orbits (TBD)	C-Band radar transmitter and onboard AIS receiver at 500 km orbital altitude.	Deployed in 2019	Svalbard (Norway)
ICEYE	18	6 SSOs	X-band radar systems. 70 kg ICEYE-X1 satellite was launched in January 2018 to SSO at altitude 500 km - the first of multiple spacecraft that will go up in the coming years.	9 satellites will be launched by the end of 2019	---

Table 5 – Hybrid-sensors (Optical and SAR) constellations.

Mission	# Sats	# Orbits & type	Constellation configuration comments	Operational status	PGS
KOMPSAT (Earth-i)	4	3 SSOs	4 satellites each in a single orbit. 2 Optical satellites in the same plane but in different altitudes KOMPSAT 2 at and KOMPSAT 3. An optical one on a single plane, SSO at altitude 528 km (KOMPSAT 3A). The fourth KOMPSAT 5 on SSO dawn-dusk at altitude 550 km.	Operational	Korea Aerospace Research Institute (KARI) GS
Copernicus program (Sentinels)	7	4 SSOs	Non-uniform distribution although every two identical pairs is equally spaced in the same orbit. SENTINEL-1A and 1B are identical SAR satellites on the same SSO orbit equally spaced at altitude 695 km and inclination 98.18°. SENTINEL-2A and 2B are identical high-resolution optical satellites on the same SSO orbit equally spaced at altitude 789.8 km and inclination 98.56°. SENTINEL-3A and 3B same SSO orbit plane at inclination 98.62° but at slightly different altitudes, 804 km and 814, respectively. They have two types of payloads (optical payload, topographic payload) to provide data for services relevant to the ocean and land. SENTINEL-5P on a single SSO at inclination 98.73°.	Operational	GEO relay by optical communication
Airbus Defence and Space Constellation	7	3 SSOs	Non-uniform distribution. 3 SAR satellites and 4 optical. TANDEM-X and TERRASAR-X fly in a formation flying (FF) just a few hundred meters apart and are joined by PAZ satellite in 2018. These SAR satellites non-uniformly distributed in the same dawn-dusk SSO at 505 km altitude. PLEIADES 1A and PLEIADES 1B are uniformly distributed in the same SSO at 694 km altitude. SPOT 6 and 7 are uniformly distributed on a single SSO at altitude 694 km. The four optical satellites are virtually 90° apart.	2007-2014 Operational	Astrium network Nearly 40 Direct Receiving Stations (DRS)
OptiSAR	16	2 orbits, SSO and MIO	The constellation is formed of 8 optical satellites and 8 SAR satellites at an altitude 450 km split across two orbital planes: SSO and Mid-Inclination Orbit (MIO) at inclination 45°. The satellites are equally distributed in each plane, with SAR satellite flying one to two minutes ahead of an optical satellite, which equipped with a high-resolution multispectral sensor.	Under development	---

Table 6 – Representative classes of RSSCs.

Inclination # of planes and altitude	Sun Synchronous Orbit (SSO)	Inclined orbit	Hybrid
Same plane Same altitude	Class # 1 RapidEye, RADARSAT, COSMO-SkyMed (1 st & 2 nd generations), and Dove satellites (Flock-1C, Flock-3P, Flock-3M, Flock-3P', Flock-2K, Flock-2P, Flock-3R)	Class # 2 Flock 2e (51.6°), Flock 2e' (51.6°), and NASA's CYGNSS (35°)	---
Different planes Same altitude	Class # 3 Vivid-I, SAR-Lupe, and ICEYE	Class # 4 XpressSAR, MicroSAR, and FormoSat-3	Class # 5 OptiSAR and NovaSAR
Different planes Different altitude	Class # 6 DMC, SkySat constellation, Digital-Globe, KOMPSAT, Copernicus (Sentinels), and Airbus	---	Class # 7 Lemur-2 Constellation of Spire Global, FormoSat-7, and CICERO

5. CONCLUSIONS

In recent years, heterogeneous satellite constellations that use ISLs between two different functional constellations, a RSSC dedicated to EOs and a RCSC dedicated to command delivery and relay data back to Earth, have attracted considerable interest. The design of general-purpose RCSCs that serve a range of RSSCs is complicated by the many different RSSCs that have been deployed to date and the effort required to test the performance of a candidate RCSC against each one. Multiple candidate configurations of RCSCs may be appropriate for servicing the same RSSC given the coverage requirements of its configuration. Consequently, to find out the optimum configuration of a heterogeneous constellation, it is necessary to study several constellation combinations.

Previous progress in this area has been hampered because of the large size of the solution space. This necessitates a classification for the RSSCs configurations to reduce the solution space. Previous work in this field has focused on categorizing the remote sensing satellites based on their mission application or sensor type. However, they did not introduce a classification to the configurations of RSSCs. Accordingly, we have sought to determine whether a pattern in RSSCs exists that would allow us to represent the greater population by a much smaller set of representative classes.

Based on a statistical analysis of NORAD TLEs for 34 RSSCs and a thorough review of their missions, we propose a nine-class solution based upon orbital inclination (three categories) and a number of planes and different altitudes (three categories) although only seven of these classes are populated by actual RSSCs. The actual number of planes, orbital altitudes, and orbital inclinations are represented by

simple distributions. Moreover, we obtained the major RSSC parameters and their range of values where a combination of these parameters can be used for forming representative configurations of RSSCs. The results allow the performance of RCSCs to be tested against a broad and representative set of RSSCs with far less effort than testing against an exhaustive set and far more confidently than against a random set. Finally, this paper presents two major outcomes:

1. Identifying nine-class representative configurations, which are useful when conducting any related simulation-based studies.
2. Obtaining the major RSSC parameters and their range of values where a combination of these parameters can be used for forming representative configurations of RSSCs.

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